

Answer all questions.

1. Absorption spectra can be used to provide information about stars and galaxies. The diagram below shows a simplified absorption spectrum from our Sun.



- (a) Place a tick (✓) in the box next to the **three** correct statements below. [3]

- The dark lines in the spectra from distant galaxies are blue shifted. ☐
- The further away a galaxy is the longer the wavelength of the dark lines. ☐
- The dark lines in the spectra from distant galaxies are green shifted. ☐
- The dark lines can be used to identify the elements present in the star / galaxy. ☐
- The dark lines in the spectra from distant galaxies are red shifted. ☐
- The further away a galaxy is the shorter the wavelength of the dark lines. ☐

- (b) The shift of the absorption lines in spectra from distant galaxies provides evidence for the Big Bang model of the creation of the Universe. State another piece of evidence supporting the Big Bang. [1]

.....

4

3420U201
03

